

Muhammad Bilal

AI Engineer — LLM Applications, Agentic Systems & RAG Pipelines
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Profile Summary

Muhammad Bilal is an AI Engineer with hands-on experience building LLM-powered applications, LangChain-based agentic systems, and Retrieval-Augmented Generation (RAG) pipelines. He is proficient in Python and FastAPI, and focuses on developing and deploying scalable AI APIs and production-ready multi-agent backends.

Technical Skills

Programming and Backend: Muhammad Bilal's skills in programming and backend include Python, FastAPI, REST APIs, Async Programming, Docker, SQLAlchemy.

Agentic AI and LLM: Muhammad Bilal's skills in agentic ai and llm include LangChain, LangGraph, LangSmith, RAG, Agentic AI, Groq, Hugging Face, FAISS.

Machine Learning and Data Science: Muhammad Bilal's skills in machine learning and data science include Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, TF-IDF.

Databases: Muhammad Bilal's skills in databases include MongoDB, SQLite, PostgreSQL, MySQL.

Security and DevOps: Muhammad Bilal's skills in security and devops include JWT, Argon2, GitHub, HashiCorp Vault.

Computer Vision: Muhammad Bilal's skills in computer vision include MTCNN, InsightFace, Cosine Similarity.

Professional Experience

AI Engineer Intern — Optimum Tech, Faisalabad

Muhammad Bilal completed an internship at Optimum Tech in Faisalabad, working as an AI Engineer Intern. During the internship, he learned and applied LangChain, RAG pipelines, chatbot development, and FastAPI for building AI-powered applications.

Freelance Data Scientist

Muhammad Bilal has worked as a freelance Data Scientist, handling data analysis and machine learning projects for clients on a freelance basis.

Projects

AI Research Agent

Muhammad Bilal built a five-agent self-correcting pipeline using Groq (Llama-3.3-70B-Versatile) and LangChain, in which a Critic Agent scored and automatically regenerated reports up to two times before final delivery. He developed a RAG-based PDF question-answering system with Tavily live search and TF-IDF retrieval, and served it through FastAPI REST endpoints secured with JWT authentication and Argon2. He converted natural language queries into MongoDB queries using the Google Gemini API, supporting AND/OR filters, comparison operators, and regex search. He deployed LLM-powered applications using FastAPI and Docker to ensure scalable and consistent production-ready deployments.

AI Mongo Query Agent

Muhammad Bilal converted natural language into MongoDB queries using the Google Gemini API, supporting AND/OR filters, comparison operators, and regex search. He deployed the application via FastAPI and Docker for scalable, consistent, production-ready deployment.

Face Recognition Attendance System

Muhammad Bilal built a real-time attendance pipeline using MTCNN for face detection and InsightFace (buffalo_l) for 512-dimensional embedding extraction, matched via Cosine Similarity with a 0.6 threshold over WebSocket streams. He implemented per-class MongoDB isolation with dynamic students-{class} and attendance-{class} collections, ensuring zero-leakage multi-tenancy across departments. He developed a FastAPI REST and WebSocket backend with JWT authentication, Excel attendance export, and batch image-based attendance marking.

MediMind — Medical Chatbot

Muhammad Bilal built a RAG pipeline using LangChain, the Groq API, and Hugging Face embeddings with FAISS for precise medical PDF retrieval. He designed a Streamlit user interface with session memory for natural conversation flow, monitored through LangSmith.

Household Energy Demand Forecasting

Muhammad Bilal built a time-series forecasting pipeline on a dataset of more than two million records, using feature engineering, exploratory data analysis, and hypothesis testing to identify peak demand patterns. He trained and compared Ridge, Random Forest, XGBoost, and Gradient Boosting models, achieving an R-squared of 0.9999 with Gradient Boosting.

Education

Muhammad Bilal is pursuing a BS in Computer Science at National Textile University, Faisalabad, from 2022 to 2026. He completed his Intermediate in Computer Science at Government Graduate College, Samundari, from 2020 to 2022.

Certificates

Muhammad Bilal holds a Data Science certificate from Code With Harry, covering data analysis, machine learning fundamentals, Pandas, and Scikit-learn. He also received a Certificate of Participation in DECTHON 2025, an online Artificial Intelligence and Machine Learning hackathon organized by Decentral

Developers, dated 01-02-2025.